

1.5

Second-order Equation $\Rightarrow$ Two First-order equations

$$x' = \frac{dx}{dt}, \quad \ddot{x} = \frac{d^2x}{dt^2}, \quad x(t_0) = x_0, \quad \dot{x}(t_0) = v_0$$

$$v(t) = \dot{x}(t)$$

$$\dot{x} = v, \quad \dot{v} = f(t, x, v)$$

Euler's method in 2 dimensions

$$\begin{aligned} x(t_{n+1}) &\approx x(t_n) + \dot{x}(t_n) \Delta t \\ v(t_{n+1}) &\approx v(t_n) + \dot{v}(t_n) \Delta t \end{aligned} \Rightarrow \begin{aligned} x_{n+1} &\approx x_n + v_n \Delta t \\ v_{n+1} &\approx v_n + f(t_n, x_n, v_n) \Delta t \end{aligned}$$

$$e_n = \frac{x_n - x}{x}, \quad \frac{x(t_n) - x(t_n)}{x(t_n)}$$

$$\frac{x(t_n) - x(t_n)}{x(t_n)} = \frac{x(t_n) - x(t_n)}{x(t_n)}$$

The first step error	$(x_1, x_2) = (1, -0.25)$	$(x_2, x_3) = (1.1, -0.25)$	$(x_3, x_4) = (1.2, -0.25)$	$(x_4, x_5) = (1.3, -0.25)$
$\frac{x(t_1) - x(t_1)}{x(t_1)}$	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000
$\frac{x(t_2) - x(t_2)}{x(t_2)}$	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000
$\frac{x(t_3) - x(t_3)}{x(t_3)}$	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000
$\frac{x(t_4) - x(t_4)}{x(t_4)}$	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000

The error between $f(t, x, v)$ and Euler's approx. converges as $h \rightarrow 0$.

The Linear Second-order equation

$$x'' + p(x) \dot{x} + q(x) x = f(x), \quad x(0) = x_0, \quad \dot{x}(0) = v_0$$

Remark: p, q is homogeneous when $f(x) = 0$.

Superposition

for $x_1, x_2, \dots, x_n$ suppose x_1, x_2 both solve this EA

Superposition: for any constants $c_1, c_2, \dots, c_n$ $\sum c_i x_i$ is also a solution

$$p(x) = \sum_{i=1}^n c_i x_i(t) = \sum_{i=1}^n c_i (x_i(t) + \dot{x}_i(t) + x_i(t)) = \sum_{i=1}^n c_i (x_i(t) + \dot{x}_i(t) + x_i(t))$$

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The Wronskian and initial conditions

$$W(x_1, x_2) = \begin{vmatrix} x_1(t) & x_2(t) \\ \dot{x}_1(t) & \dot{x}_2(t) \end{vmatrix}$$

$$\Rightarrow \frac{d}{dt} W(x_1, x_2) = 0$$

$$W(x_1, x_2) = \begin{vmatrix} x_1(t) & x_2(t) \\ \dot{x}_1(t) & \dot{x}_2(t) \end{vmatrix} = \begin{vmatrix} x_1(t) & x_2(t) \\ \dot{x}_1(t) & \dot{x}_2(t) \end{vmatrix}$$

If $W(x_1, x_2) \neq 0$, the initial condition system has a unique solution

$$c_1 x_1(t) + c_2 x_2(t) = x(t), \quad \dot{c}_1 x_1(t) + \dot{c}_2 x_2(t) = \dot{x}(t)$$

$x(t) = c_1 x_1(t) + c_2 x_2(t)$ is the general solution

The Wronskian can't be constant, but if nonzero at one point it is nonzero throughout the interval. Testing at $t=0$ is often convenient.

Homogeneous linear second-order ODEs

$$x'' + p(x) \dot{x} + q(x) x = 0, \quad x(0) = x_0, \quad \dot{x}(0) = v_0$$

$$T(x) = x'' + p(x) \dot{x} + q(x) x = 0$$

$$x'' + p(x) \dot{x} + q(x) x = 0, \quad x(0) = x_0, \quad \dot{x}(0) = v_0$$

$$\Rightarrow x'' + p(x) \dot{x} + q(x) x = 0$$

$$r^2 + pr + q = 0 \Rightarrow \begin{cases} \text{two distinct real roots: this lecture} \\ \text{two complex conjugate roots: next lecture} \\ \text{repeated root: next lecture} \end{cases}$$

For $r_1 \neq r_2$, real (2 distinct roots)

$$x(t) = c_1 e^{r_1 t} + c_2 e^{r_2 t}$$

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1.6

Complex conjugate roots

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